

Francisco Lopes

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EDUCATION

University of Minho

MSc Software Engineering

Braga, Portugal

Sep 2025 – Present

University of Trás-os-Montes and Alto Douro

BSc Software Engineering

Vila Real, Portugal

Sep 2022 – Jun 2025

EXPERIENCE

Software Engineer Intern

Neadvance

Aug 2024 – Sep 2024

Braga, Portugal

- Identified a workflow bottleneck where large Git pushes were difficult to review, and independently designed and built a Git diff visualisation tool from scratch using Python
- Developed a clean interface that parses and displays code changes with clear indentation, highlighting added and removed lines for rapid review
- Delivered a measurable improvement in the team's code review efficiency, with the tool adopted and continued in active use after the internship ended, working within the team's Git branching and pull request workflow

PROJECTS

Distributed IoT Telemetry Platform | Java, Erlang, Java NIO, RxJava, ZeroMQ

Feb 2026 – Jul 2026

- Designed and implemented concurrent message handling using Java NIO and RxJava reactive streams; ZMQ DEALER/ROUTER for scalable async inter-server communication
- Gained deep practical understanding of distributed coordination, message-passing protocols, and fault-tolerant system design
- MSc Distributed Systems course project — Grade: 18/20

Votium — Blockchain Voting Application | TypeScript, JavaScript, HTML,

CSS, React, Solidity, MongoDB

Feb 2025 – Jul 2025

- Architected and deployed smart contracts in Solidity on an Ethereum-compatible (Ganache) network, designing on-chain data structures for vote storage and verification
- Developed the backend with MongoDB to manage application data and integrate off-chain services with the blockchain
- Built the frontend using React and TypeScript, implementing real-time blockchain event listening and interactions with smart contracts
- BSc Final Project — Grade: 17/20

Automated Cloud Deployment & Optimization of AirTrail | Python,

Kubernetes, Ansible, Locust

Sep 2025 – Dec 2025

- Automated the full deployment lifecycle of a web application on Google Kubernetes Engine using Ansible, covering cluster provisioning, storage, networking, and health checks
- Load-tested the platform with Locust and used cloud monitoring data to identify and eliminate a scalability bottleneck through autoscaling, cutting the error rate from 21% to 4% and halving latency under load
- MSc Cloud Computing course project — Grade: 17.5/20

TECHNICAL SKILLS

Languages: Java, Python, Erlang, TypeScript, JavaScript, Bash

Frameworks & Tools: Git, ZeroMQ, RxJava, Java NIO, React, Node.js, MongoDB

Cloud & Infrastructure: Kubernetes, Docker, Ansible, Google Cloud Monitoring, Locust, Microservices Architecture, Container Architecture, Infrastructure as Code

Domains: Backend Engineering, Distributed Systems, Cloud & DevOps, Event-Driven Systems, Infrastructure Automation, Software Architecture, Blockchain

LANGUAGES

Portuguese: Native | **English:** Fluent (written and oral)